OAVS • TGT COMPUTER TEACHER EXAMINATION

11 Distributed Databases

11.1 Concepts, Data Distribution Techniques

Exam	OAVS TGT Computer Teacher Recruitment Examination
Chapter	11. Distributed Databases
Topic Covered	11.1 Concepts, Data Distribution Techniques
Question Type	Multiple Choice Questions (MCQ) with Answer & Explanation
Total Questions	79 high-value, exam-oriented, non-repetitive MCQs
Sections	8 focused sections (Concepts, Data Distribution, Techniques, Scenario/Diagram, Difficult & Tricky, Mixed Practice, Additional High-Value, Allocation & Catalog)

This question bank is strictly limited to the syllabus topic **"11. Distributed Databases — 11.1 Concepts, Data Distribution Techniques."** It excludes distributed transaction management, concurrency control, recovery, distributed query processing, and any advanced/NoSQL/Big Data/Cloud topics not covered under this section.

1. Distributed Database Concepts	**10 Qs**
2. Data Distribution Concepts	**8 Qs**
3. Data Distribution Techniques	**10 Qs**
4. Scenario / Diagram-Based Questions	**8 Qs**
5. Difficult & Tricky Questions	**8 Qs**
6. Mixed OAVS TGT Practice	**10 Qs**
7. Additional High-Value Questions	**15 Qs**
8. Data Allocation & Catalog Questions	**10 Qs**

11 Distributed Databases

11.1 Concepts, Data Distribution Techniques — MCQ Question Bank

1. Distributed Database Concepts

Q1. What is a Distributed Database (DDB)?

A. A collection of multiple, logically related databases distributed over a computer network

B. A single database stored on one central server

C. A database that only supports SQL queries

D. A backup copy of a centralized database

Ans: A — A collection of multiple, logically related databases distributed over a computer network

Explanation: A DDB is a logically integrated collection of databases physically distributed across multiple sites connected by a network.

Q2. Which of the following is NOT a characteristic of a distributed database?

A. Data is stored at multiple physical locations

B. All data must reside in a single location

C. Sites are connected through a communication network

D. Each site can have local autonomy

Ans: B — All data must reside in a single location

Explanation: Storing all data at a single location is a feature of a centralized database, not a distributed one.

Q3. In a distributed database, "site autonomy" refers to:

A. Complete independence of a site from the network

B. The ability of a local site to manage its own database independently

C. The main server controlling all sites

D. A site that has no data of its own

Ans: B — The ability of a local site to manage its own database independently

Explanation: Site autonomy means each local site can manage its own database and operations without depending entirely on a central controller.

Q4. Which architecture involves multiple servers cooperating as equals, each contributing data and processing power?

A. Client-Server architecture

B. Master-Slave architecture

C. Centralized architecture

D. Peer-to-Peer architecture

Ans: D — Peer-to-Peer architecture

Explanation: In peer-to-peer distributed architecture, all sites act as equal partners, sharing data and processing responsibilities.

Q5. A distributed database in which all sites use the same DBMS software is called:

A. Heterogeneous distributed database

B. Hybrid distributed database

C. Homogeneous distributed database

D. Federated database

Ans: C — Homogeneous distributed database

Explanation: A homogeneous distributed database uses identical DBMS software and schema structure across all sites.

Q6. Which of the following is a primary reason organizations adopt distributed databases?

A. To store data at a single site for simplicity

B. To eliminate the need for a network

C. To reflect the organizational structure and improve local data access

D. To reduce the number of database users

Ans: C — To reflect the organizational structure and improve local data access

Explanation: Distributed databases are often designed to mirror organizational structure, allowing faster local access and autonomy for each branch/unit.

Q7. Statement 1: A distributed database can improve reliability since failure of one site does not necessarily stop operations at other sites.
Statement 2: A distributed database always requires less hardware than a centralized database.

A. Both statements are true

B. Only Statement 2 is true

C. Both statements are false

D. Only Statement 1 is true

Ans: D — Only Statement 1 is true

Explanation: Distributed systems improve fault tolerance (Statement 1), but they typically require more hardware/network resources, not less.

Q8. Assertion (A): Distributed databases increase system reliability and availability.
Reason (R): In a distributed database, data is duplicated or spread across multiple independent sites, so failure of one site does not necessarily cause total system failure.

A. Both A and R are true, but R does not explain A

B. A is true, R is false

C. Both A and R are true, and R correctly explains A

D. A is false, R is true

Ans: C — Both A and R are true, and R correctly explains A

Explanation: Distribution and replication of data across independent sites directly improve reliability and availability, correctly justifying the assertion.

Q9. Match the following architecture types with their description:
1. Client-Server
2. Peer-to-Peer
a. Multiple servers act as equal partners
b. Clients send requests, server manages and processes data

A. 1-a, 2-b

B. 1-a, 2-a

C. 1-b, 2-a

D. 1-b, 2-b

Ans: C — 1-b, 2-a

Explanation: In Client-Server, clients request services from a central server; in Peer-to-Peer, all sites participate equally.

Q10. Which of the following BEST distinguishes a distributed database from simply "a network of independent databases"?

A. A distributed database has no network connection

B. A distributed database appears to users as a single, logically integrated database despite physical distribution

C. A distributed database is always heterogeneous

D. A distributed database cannot have local databases

Ans: B — A distributed database appears to users as a single, logically integrated database despite physical distribution

Explanation: The defining feature of a DDB is logical integration: users interact with it as one database even though data is physically spread out.

2. Data Distribution Concepts

Q11. In the context of distributed databases, "data distribution" refers to:

A. The process of dividing and placing data across multiple sites in a network

B. Deleting duplicate data from a single server

C. Compressing data to save space

D. Encrypting data before storage

Ans: A — The process of dividing and placing data across multiple sites in a network

Explanation: Data distribution is the process of dividing a database and storing/placing its parts across different network sites.

Q12. Which term describes dividing a database into smaller logical pieces for distribution across sites?

A. Normalization

B. Indexing

C. Encryption

D. Fragmentation

Ans: D — Fragmentation

Explanation: Fragmentation is the process of dividing a database into fragments (pieces) that are distributed across different sites.

Q13. Storing multiple copies of the same data at different sites is known as:

A. Fragmentation

B. Allocation

C. Partitioning

D. Replication

Ans: D — Replication

Explanation: Replication means maintaining duplicate copies of data at more than one site to improve availability and access speed.

Q14. The process of deciding where each fragment or copy of data will be physically stored is called:

A. Replication

B. Data Allocation

C. Query Optimization

D. Indexing

Ans: B — Data Allocation

Explanation: Data allocation is the decision process of assigning fragments (fragmented or replicated) to specific sites in the network.

Q15. In "full replication," a distributed database:

A. Stores a complete copy of the entire database at every site
B. Stores each data item at exactly one site
C. Never duplicates any data
D. Divides data into fragments without copying

Ans: A — Stores a complete copy of the entire database at every site

Explanation: Full replication means the entire database is duplicated and maintained at every site in the network.

Q16. Which data distribution strategy stores each data item at only a single site with no duplication?

A. Full replication
B. Partial replication
C. No replication (Partitioned allocation)
D. Mixed replication

Ans: C — No replication (Partitioned allocation)

Explanation: Under a "no replication" or fully partitioned strategy, each item exists at exactly one site, with no duplicate copies.

Q17. Statement 1: Replication improves data availability but increases storage and update overhead.
Statement 2: Fragmentation always requires making multiple copies of the same data.

A. Both statements are true
B. Only Statement 1 is true
C. Only Statement 2 is true
D. Both statements are false

Ans: B — Only Statement 1 is true

Explanation: Replication trades storage/update cost for availability, but fragmentation divides data into distinct pieces; no copies are needed.

Q18. Assertion (A): Data distribution concepts include fragmentation, replication, and allocation.
Reason (R): These three concepts together determine how a database's data is divided, duplicated, and placed across the sites of a distributed system.

A. Both A and R are true, but R does not explain A
B. Both A and R are true, and R correctly explains A
C. A is true, R is false
D. A is false, R is true

Ans: B — Both A and R are true, and R correctly explains A

Explanation: Fragmentation, replication, and allocation are the three core building blocks of data distribution design in distributed databases.

3. Data Distribution Techniques

Q19. Horizontal fragmentation divides a relation (table) based on:

A. Columns (attributes)
B. File size
C. Index values only
D. Rows (tuples) based on a condition

Ans: D — Rows (tuples) based on a condition

Explanation: Horizontal fragmentation splits a table into subsets of rows, each satisfying a specified selection condition.

Q20. Vertical fragmentation divides a relation (table) based on:

A. Columns (attributes), typically keeping a common key in each fragment

B. Rows satisfying a condition

C. Network bandwidth

D. Number of users accessing it

Ans: A — Columns (attributes), typically keeping a common key in each fragment

Explanation: Vertical fragmentation splits a table by columns, with the primary key replicated in every fragment to allow reconstruction.

Q21. A fragmentation technique that combines both row-based and column-based division is called:

A. Horizontal fragmentation

B. Mixed (Hybrid) fragmentation

C. Vertical fragmentation

D. Full replication

Ans: B — Mixed (Hybrid) fragmentation

Explanation: Mixed or hybrid fragmentation applies horizontal and vertical fragmentation together, dividing a relation by both rows and columns.

Q22. The "completeness" rule of fragmentation states that:

A. No two fragments may share any data item

B. Fragments must always be stored at a single site

C. Every data item of the original relation must appear in at least one fragment

D. Every fragment must have the same number of rows

Ans: C — Every data item of the original relation must appear in at least one fragment

Explanation: Completeness ensures no data is lost during fragmentation: every item must map into at least one fragment.

Q23. The "reconstruction" rule of fragmentation ensures that:

A. Fragments can never be combined again

B. The original relation can always be recreated from its fragments

C. Each fragment must be replicated at every site

D. Fragments must be encrypted before storage

Ans: B — The original relation can always be recreated from its fragments

Explanation: The reconstruction rule guarantees a relational operator (such as union or join) exists that can rebuild the original relation from its fragments.

Q24. The "disjointness" rule of fragmentation requires that:

A. Every fragment overlaps with every other fragment

B. All fragments be stored at the same site

C. Data items generally should not be duplicated across horizontal fragments

D. Fragments cannot be reconstructed

Ans: C — Data items generally should not be duplicated across horizontal fragments

Explanation: Disjointness means (for horizontal fragmentation) that each data item belongs to only one fragment, avoiding unnecessary duplication.

Q25. To reconstruct a relation that has been horizontally fragmented, which relational operation is typically used?

A. Union
B. Join
C. Division
D. Cartesian product

Ans: A — Union

Explanation: Horizontal fragments contain disjoint sets of rows with identical structure, so a UNION recombines them into the original relation.

Q26. To reconstruct a relation that has been vertically fragmented, which relational operation is typically used?

A. Union
B. Intersection
C. Selection
D. Join (using the common key)

Ans: D — Join (using the common key)

Explanation: Vertical fragments share a common key attribute, so a JOIN on that key is used to reconstruct the original relation.

Q27. Which of the following is essential in every vertical fragment to allow later reconstruction?

A. The primary key of the original relation
B. A foreign key from another table
C. An index on all columns
D. A backup copy of the whole table

Ans: A — The primary key of the original relation

Explanation: The primary key must be included in each vertical fragment so the fragments can be joined back together correctly.

Q28. Match the following fragmentation types with their basis of division:
1. Horizontal
2. Vertical
3. Mixed
a. Divides by attributes/columns
b. Divides by rows/tuples
c. Divides by both rows and columns

A. 1-b, 2-a, 3-c
B. 1-a, 2-b, 3-c
C. 1-c, 2-a, 3-b
D. 1-b, 2-c, 3-a

Ans: A — 1-b, 2-a, 3-c

Explanation: Horizontal fragmentation divides rows, vertical fragmentation divides columns, and mixed fragmentation applies both.

4. Scenario / Diagram-Based Questions

Q29. A bank has branches in Bhubaneswar, Rourkela, and Cuttack. Each branch stores only the records of customers belonging to that branch. This is an example of:

A. Vertical fragmentation
B. Horizontal fragmentation
C. Full replication
D. Mixed fragmentation

Ans: B — Horizontal fragmentation

Explanation: Splitting customer rows by branch location (a condition on rows) is a classic horizontal fragmentation scenario.

Q30. An organization splits its EMPLOYEE table so that the Payroll department stores salary-related columns and the HR department stores personal-detail columns, with EmpID common to both. This is an example of:

A. Vertical fragmentation
B. Horizontal fragmentation
C. Replication
D. Allocation only

Ans: A — Vertical fragmentation

Explanation: Dividing a table by columns (attributes) while retaining the key (EmpID) in each part is vertical fragmentation.

Q31. In a basic distributed database architecture diagram, each "site" typically consists of:

A. Only a network cable
B. A local database, a local DBMS, and a communication component connecting it to other sites
C. Only a central server with no local storage
D. A single user terminal with no software

Ans: B — A local database, a local DBMS, and a communication component connecting it to other sites

Explanation: A site in a DDB architecture includes its own local database and DBMS, plus a communication interface linking it to the other sites.

Q32. A university keeps identical copies of its central course-catalog database at every campus so students can access it even if the main server is down. This illustrates:

A. Horizontal fragmentation
B. Full replication
C. Vertical fragmentation
D. Partitioned allocation

Ans: B — Full replication

Explanation: Storing an identical complete copy of the same data at every site is full replication, chosen here for availability.

Q33. A retail chain divides its SALES table so that each store keeps only the rows for transactions made at that store, while all stores share the same table structure. Which distribution technique is used?

A. Horizontal fragmentation
B. Vertical fragmentation
C. Mixed fragmentation
D. Full replication

Ans: A — Horizontal fragmentation

Explanation: Each store holding only its own transaction rows, with the same schema, is horizontal fragmentation based on store location.

Q34. Consider a PRODUCT table horizontally fragmented into F1 (Price > 1000) and F2 (Price ≤ 1000). If a product with Price = 1000 appears in BOTH F1 and F2, which fragmentation rule is violated?

A. Completeness
B. Reconstruction
C. Disjointness
D. Allocation

Ans: C — Disjointness

Explanation: Disjointness is violated because the same data item (the ₹1000 product) appears in more than one fragment.

Q35. A company wants each fragment of a table to be recreated back into the full original table whenever required. Which fragmentation rule guarantees this is possible?

A. Completeness
B. Disjointness
C. Replication
D. Reconstruction

Ans: D — Reconstruction

Explanation: The reconstruction rule ensures the original relation can always be rebuilt from its fragments using relational operators.

Q36. A simple two-site diagram shows Site A and Site B connected by a network, each with its own local DBMS. If Site A fails, Site B still functions independently. This demonstrates which benefit of distributed databases?

A. Improved reliability/availability through site independence
B. Reduced hardware cost
C. Elimination of the need for a network
D. Guaranteed homogeneous software at both sites

Ans: A — Improved reliability/availability through site independence

Explanation: Because each site can operate independently, failure at one site does not stop operations at the other, improving overall reliability.

5. Difficult & Tricky Questions

Q37. Which of the following is a drawback of full replication as a data distribution technique?

A. It reduces data availability
B. It prevents any site from accessing data locally
C. It increases storage requirements and the overhead of keeping all copies updated
D. It makes reconstruction of the original relation impossible

Ans: C — It increases storage requirements and the overhead of keeping all copies updated

Explanation: Although full replication boosts availability, maintaining identical copies at every site consumes more storage and requires updating every copy.

Q38. Which of the following correctly ranks the general storage cost of distribution strategies from LOWEST to HIGHEST?

A. Full replication < Partial replication < No replication
B. No replication < Partial replication < Full replication
C. Partial replication < No replication < Full replication
D. Full replication < No replication < Partial replication

Ans: B — No replication < Partial replication < Full replication

Explanation: No replication (data stored once) uses the least storage, partial replication uses more, and full replication (a complete copy everywhere) uses the most.

Q39. Assertion (A): Vertical fragmentation of a relation without including the primary key in each fragment makes reconstruction impossible.
Reason (R): Reconstruction of vertically fragmented data depends on joining fragments using a common key attribute.

A. Both A and R are true, and R correctly explains A
B. Both A and R are true, but R does not explain A
C. A is true, R is false
D. A is false, R is true

Ans: A — Both A and R are true, and R correctly explains A

Explanation: Without a shared key, there is no attribute to join the fragments on, so the original table cannot be reconstructed.

Q40. A designer must choose fragmentation for a CUSTOMER table where regional offices need only their own region's customers, but every office needs every column. Which technique fits best?

A. Horizontal fragmentation

B. Vertical fragmentation

C. Full replication

D. No fragmentation, only replication of the primary key

Ans: A — Horizontal fragmentation

Explanation: Since the requirement is dividing by region (rows) while keeping all columns, horizontal fragmentation is appropriate.

Q41. Which of the following statements is TRUE regarding mixed (hybrid) fragmentation?

A. It can only be applied to tables with a single column

B. It is obtained by first applying one fragmentation type and then applying the other type on the resulting fragments

C. It eliminates the need for the completeness rule

D. It is identical to full replication

Ans: B — It is obtained by first applying one fragmentation type and then applying the other type on the resulting fragments

Explanation: Mixed fragmentation is typically achieved by applying horizontal fragmentation and then vertically fragmenting the results (or vice versa).

Q42. In a distributed database, if data is partitioned (no replication) rather than replicated, what is a likely TRADE-OFF?

A. Higher storage cost but lower availability during site failure

B. No difference in availability compared to full replication

C. Guaranteed availability regardless of site failure

D. Lower storage cost but reduced availability if the site holding that data fails

Ans: D — Lower storage cost but reduced availability if the site holding that data fails

Explanation: Partitioning saves storage since data exists only once, but if that single site fails, the data becomes unavailable elsewhere.

Q43. Which pair correctly matches a fragmentation type with the relational algebra operation it conceptually resembles?

A. Horizontal fragmentation — Projection; Vertical fragmentation — Selection

B. Horizontal fragmentation — Selection; Vertical fragmentation — Projection

C. Horizontal fragmentation — Join; Vertical fragmentation — Union

D. Horizontal fragmentation — Union; Vertical fragmentation — Join

Ans: B — Horizontal fragmentation — Selection; Vertical fragmentation — Projection

Explanation: Horizontal fragmentation resembles SELECT (choosing rows by condition); vertical fragmentation resembles PROJECT (choosing columns).

Q44. Match the following distribution concepts to their correct description:
1. Fragmentation
2. Allocation
3. Replication
a. Deciding the physical site for each fragment/copy
b. Dividing a relation into smaller pieces
c. Maintaining duplicate copies of data

A. 1-a, 2-b, 3-c
B. 1-c, 2-b, 3-a
C. 1-b, 2-c, 3-a
D. 1-b, 2-a, 3-c

Ans: D — 1-b, 2-a, 3-c

Explanation: Fragmentation divides data, allocation decides placement of fragments/copies, and replication maintains duplicate copies.

6. Mixed OAVS TGT Practice

Q45. Which of the following is NOT a basic data distribution technique used in distributed databases?

A. Horizontal fragmentation
B. Vertical fragmentation
C. Normalization
D. Replication

Ans: C — Normalization

Explanation: Normalization is a relational design technique to reduce redundancy, not a technique for placing data across sites.

Q46. A distributed database where different sites use different DBMS software and possibly different schemas is called:

A. Homogeneous distributed database
B. Heterogeneous distributed database
C. Centralized database
D. Single-site database

Ans: B — Heterogeneous distributed database

Explanation: A heterogeneous distributed database has sites running different DBMS products and/or data models, unlike a homogeneous DDB.

Q47. Which of the following is a valid purpose of using data distribution in a distributed database?

A. To make data access slower for all users
B. To force every query to visit every site
C. To support local autonomy while allowing global access to data
D. To eliminate the need for a communication network

Ans: C — To support local autonomy while allowing global access to data

Explanation: Data distribution lets each site manage its local data independently while still allowing integrated, global data access.

Q48. **Statement 1: In horizontal fragmentation, each fragment has the same schema (columns) as the original relation. Statement 2: In vertical fragmentation, the primary key is excluded from all fragments to save storage space.**

A. Only Statement 1 is true
B. Both statements are true
C. Only Statement 2 is true
D. Both statements are false

Ans: A — Only Statement 1 is true

Explanation: Horizontal fragments keep the full column structure (Statement 1 true); vertical fragments must INCLUDE the primary key (Statement 2 false).

Q49. **Which of the following best describes "location transparency" as a concept related to data distribution?**

A. Users must know the exact physical site where data is stored to access it
B. Data cannot be accessed from more than one location
C. All data must be stored at the user's local site only
D. Users can access data without needing to know its physical storage location

Ans: D — Users can access data without needing to know its physical storage location

Explanation: Location transparency lets users query data without knowing which site physically holds it; the DDBMS handles the location.

Q50. **Assertion (A): A distributed database is generally more complex to design than a centralized database. Reason (R): Data distribution requires additional decisions about fragmentation, replication, and allocation across multiple sites.**

A. Both A and R are true, but R does not explain A
B. A is true, R is false
C. Both A and R are true, and R correctly explains A
D. A is false, R is true

Ans: C — Both A and R are true, and R correctly explains A

Explanation: The added design decisions of fragmentation, replication, and allocation are what make DDB design more complex.

Q51. **An organization with offices in three cities wants EACH office to store data for ALL customers (not just its own), fully duplicated, so any office can serve any customer. The required technique is:**

A. Horizontal fragmentation
B. Vertical fragmentation
C. Full replication
D. No distribution technique

Ans: C — Full replication

Explanation: Storing a complete duplicate copy of the entire customer data at every office is a direct application of full replication.

Q52. **Which of the following is generally considered a benefit of properly designed data distribution in a DDBMS?**

A. Increased single point of failure
B. Reduced local data availability
C. Mandatory use of only one DBMS vendor
D. Faster local access to frequently used data and improved fault tolerance

Ans: D — Faster local access to frequently used data and improved fault tolerance

Explanation: Well-designed distribution places data close to where it is most used, giving faster local access and less reliance on any single site.

Q53. Match the following terms with their correct definitions:
1. Homogeneous DDB
2. Heterogeneous DDB
3. Site autonomy
a. All sites run the same DBMS software
b. Sites may run different DBMS software/schemas
c. Local site manages its own database independently

A. 1-b, 2-a, 3-c
B. 1-a, 2-c, 3-b
C. 1-a, 2-b, 3-c
D. 1-c, 2-a, 3-b

Ans: C — 1-a, 2-b, 3-c

Explanation: Homogeneous DDBs share identical DBMS software; heterogeneous DDBs may differ across sites; site autonomy means independent local management.

Q54. In distributed database design, which of the following correctly orders the typical design steps?

A. Allocation → Fragmentation → Query processing
B. Fragmentation → Allocation → Replication decision
C. Replication → Normalization → Fragmentation
D. Indexing → Fragmentation → Allocation

Ans: B — Fragmentation → Allocation → Replication decision

Explanation: A relation is first fragmented, then replication is decided, and finally fragments/copies are allocated to sites.

7. Additional High-Value Questions

Q55. Which of the following correctly identifies the TWO core strategies used to distribute data across sites in a DDB?

A. Fragmentation and Replication
B. Normalization and Denormalization
C. Encryption and Hashing
D. Clustering and Indexing

Ans: A — Fragmentation and Replication

Explanation: Distribution of data across sites is chiefly carried out through fragmentation (dividing data) and replication (duplicating data).

Q56. How many basic correctness rules must a valid fragmentation satisfy?

A. Two (Completeness, Reconstruction only)
B. Four (including Replication)
C. Three (Completeness, Reconstruction, Disjointness)
D. Five (including Allocation and Indexing)

Ans: C — Three (Completeness, Reconstruction, Disjointness)

Explanation: A valid fragmentation must satisfy three correctness rules: completeness, reconstruction, and disjointness.

Q57. Horizontal fragmentation is also sometimes referred to as:

A. Partitioning
B. Projection
C. Aggregation
D. Normalization

Ans: A — Partitioning

Explanation: Because it splits a table by rows using a selection condition, horizontal fragmentation is also commonly called partitioning.

Q58. Which of the following is a key DIFFERENCE between a centralized database and a distributed database?

A. A centralized database stores all data at one site; a distributed database stores data across multiple interconnected sites

B. A centralized database always has more storage capacity than a distributed one

C. A distributed database cannot be queried by end users

D. A centralized database requires a communication network while a distributed one does not

Ans: A — A centralized database stores all data at one site; a distributed database stores data across multiple interconnected sites

Explanation: The fundamental difference is the physical location of data: a single site (centralized) versus multiple networked sites (distributed).

Q59. Which of the following is generally considered a DISADVANTAGE of distributed databases?

A. Reduced local autonomy at every site

B. Increased complexity in design, management, and maintaining consistency across sites

C. Complete elimination of data redundancy

D. Inability to add new sites to the network

Ans: B — Increased complexity in design, management, and maintaining consistency across sites

Explanation: Coordinating fragmented/replicated data across multiple independent sites makes distributed databases more complex than centralized ones.

Q60. "Modular growth" as a benefit of distributed databases means:

A. The database must be redesigned completely whenever new data is added

B. Growth is only possible at the central site

C. No additional hardware can ever be added to the system

D. New sites or nodes can be added to the network without major disruption to existing sites

Ans: D — New sites or nodes can be added to the network without major disruption to existing sites

Explanation: Distributed systems allow incremental (modular) expansion: new sites can be added with minimal impact on existing sites.

Q61. Assertion (A): Reducing data redundancy is always the main goal of data distribution in a DDB.
Reason (R): Replication, a common data distribution technique, intentionally creates redundant copies of data.

A. A is false, R is true

B. Both A and R are true, and R correctly explains A

C. Both A and R are true, but R does not explain A

D. A is true, R is false

Ans: A — A is false, R is true

Explanation: Reducing redundancy is NOT the main goal of distribution (A false); replication deliberately introduces redundant copies for availability (R true).

Q62. In a distributed database, "fragmentation transparency" means:

A. Users must specify every fragment name in their query

B. Fragmentation is never used in a distributed database

C. Users can write queries without needing to know that a relation has been divided into fragments

D. Only administrators are allowed to access fragmented data

Ans: C — Users can write queries without needing to know that a relation has been divided into fragments

Explanation: Fragmentation transparency hides how a relation is split, letting users query it as if it were one whole table.

Q63. Which statement about "replication transparency" is correct?

A. It requires users to manually update every replica themselves

B. It means no replicas are allowed anywhere in the system

C. It applies only to horizontally fragmented relations

D. It allows users to be unaware of how many copies of data exist or where they are located

Ans: D — It allows users to be unaware of how many copies of data exist or where they are located

Explanation: Replication transparency hides the existence and location of multiple copies of data from the end user.

Q64. Which of the following is TRUE about the relationship between fragmentation and replication?

A. Fragmentation and replication are the same technique with different names

B. Fragmentation divides data into pieces, while replication duplicates data; a DDB design may use one, both, or neither depending on requirements

C. A distributed database must always use both fragmentation and replication together

D. Replication is simply a type of fragmentation

Ans: B — Fragmentation divides data into pieces, while replication duplicates data; a DDB design may use one, both, or neither depending on requirements

Explanation: Fragmentation and replication are distinct techniques and can be applied independently or combined based on system needs.

Q65. A company splits its ORDERS table by year (2023, 2024, 2025 orders in separate fragments at separate sites), with no duplication. Which correctness rule ensures no order record is lost during this split?

A. Disjointness

B. Reconstruction

C. Allocation

D. Completeness

Ans: D — Completeness

Explanation: Completeness guarantees that every order record from the original table is placed into at least one yearly fragment.

Q66. Which of the following is NOT a genuine advantage typically associated with distributed databases?

A. Improved local control and autonomy

B. Reflects the natural structure of the organization

C. Increased reliability and availability

D. Guaranteed zero network cost

Ans: D — Guaranteed zero network cost

Explanation: Distributed databases depend on a communication network, which has real cost; "zero network cost" is not an advantage.

Q67. Match the following transparency types with their meaning:
1. Location transparency
2. Fragmentation transparency
3. Replication transparency
a. Hides how data is split into pieces
b. Hides where data is physically stored
c. Hides the existence/number of data copies

A. 1-a, 2-b, 3-c
B. 1-b, 2-a, 3-c
C. 1-c, 2-a, 3-b
D. 1-b, 2-c, 3-a

Ans: B — 1-b, 2-a, 3-c

Explanation: Location transparency hides the physical site, fragmentation transparency hides division into pieces, and replication transparency hides duplicate copies.

Q68. Which scenario BEST illustrates the need for a distributed database rather than a centralized one?

A. A small shop wants to store its daily sales in a single spreadsheet
B. A single user wants to maintain a personal contact list
C. A standalone offline application that needs no network access
D. A multinational company wants each regional office to access and update its own local data quickly while occasionally sharing data company-wide

Ans: D — A multinational company wants each regional office to access and update its own local data quickly while occasionally sharing data company-wide

Explanation: Distributed databases suit geographically separated units that need fast local access along with occasional global data sharing.

Q69. Which data distribution outcome results from applying horizontal fragmentation on a relation and then vertically fragmenting each resulting piece?

A. A fully replicated copy of the relation at every site
B. A single unfragmented relation with no distribution
C. A mixed (hybrid) fragmentation of the original relation
D. A denormalized version of the relation

Ans: C — A mixed (hybrid) fragmentation of the original relation

Explanation: Horizontal followed by vertical fragmentation (or vice versa) produces mixed/hybrid fragmentation.

8. Data Allocation & Catalog Questions

Q70. **Match the following data allocation strategies with their description:**
1. Centralized
2. Partitioned (Fragmented)
3. Full replication
4. Partial replication
a. Entire database copy at every site
b. Whole database stored at a single site
c. Some fragments are copied to several sites, others kept at one site
d. Each fragment stored at exactly one site

A. 1-d, 2-b, 3-a, 4-c
B. 1-b, 2-d, 3-c, 4-a
C. 1-a, 2-d, 3-b, 4-c
D. 1-b, 2-d, 3-a, 4-c

Ans: D — 1-b, 2-d, 3-a, 4-c

Explanation: Centralized = whole DB at one site; partitioned = each fragment at one site; full replication = copy everywhere; partial replication = selected fragments copied.

Q71. **A relation R is horizontally fragmented into F1 and F2. F1 is stored at sites S1 and S2, while F2 is stored only at S3. Which allocation strategy is being used?**

A. Partial replication
B. Full replication
C. Centralized allocation
D. Pure partitioned allocation (no copies)

Ans: A — Partial replication

Explanation: F1 has more than one copy while F2 has a single copy, so only some fragments are replicated: partial replication.

Q72. **Which of the following SQL queries defines a HORIZONTAL fragment of the EMPLOYEE table for department number 5?**

A. SELECT Ssn, Name, Salary FROM EMPLOYEE
B. SELECT COUNT(*) FROM EMPLOYEE
C. SELECT * FROM EMPLOYEE E JOIN DEPARTMENT D ON E.Dno = D.Dno
D. SELECT * FROM EMPLOYEE WHERE Dno = 5

Ans: D — SELECT * FROM EMPLOYEE WHERE Dno = 5

Explanation: A horizontal fragment keeps all columns but only the rows satisfying a condition (Dno = 5), like a SELECT operation.

Q73. **In a distributed database, which component stores information about how relations are fragmented and where each fragment or copy is located?**

A. Global (distributed) data dictionary / catalog
B. Local index file of one site
C. Transaction log of a single site
D. Application program buffer

Ans: A — Global (distributed) data dictionary / catalog

Explanation: The global catalog (data dictionary) records fragmentation and allocation details so the DDBMS can locate data for users.

Q74. In distributed database design, the mapping of each fragment to the physical site(s) where it is stored is described by the:

A. Fragmentation schema
B. External schema
C. Allocation schema
D. Local internal schema of one site

Ans: C — Allocation schema

Explanation: The fragmentation schema defines how a relation is divided, while the allocation schema defines which site(s) store each fragment.

Q75. Why is a fragment, rather than a whole relation, usually chosen as the unit of data distribution?

A. Whole relations cannot be stored at remote sites
B. Applications usually need only subsets of a relation, so fragments can be placed near their users and reduce unnecessary data transfer
C. Fragments eliminate the need for a network
D. Fragments make reconstruction of data unnecessary

Ans: B — Applications usually need only subsets of a relation, so fragments can be placed near their users and reduce unnecessary data transfer

Explanation: Users mostly access portions of a relation, so storing fragments near their users improves local access and cuts network transfer.

Q76. The guiding principle of allocating a fragment to the site where it is accessed most frequently is based on:

A. Data locality (placing data close to its users)
B. Data normalization
C. Data encryption
D. Data redundancy elimination

Ans: A — Data locality (placing data close to its users)

Explanation: Data allocation aims at locality, so frequently used fragments are kept at or near the site that uses them most.

Q77. Consider the statements:
I. A fragment is a logical piece of a relation.
II. A fragment may be allocated to more than one site.
III. Allocation decides at which site(s) each fragment is stored.
Which statements are correct?

A. I and III only
B. II and III only
C. I, II and III
D. I and II only

Ans: C — I, II and III

Explanation: Fragments are logical pieces; placing a fragment at several sites replicates it; allocation chooses the site(s).

Q78. The software system that manages a distributed database and hides the details of distribution from users is called:

A. File Allocation Table
B. Local operating system kernel
C. Network router firmware
D. Distributed DBMS (DDBMS)

Ans: D — Distributed DBMS (DDBMS)

Explanation: A DDBMS manages the distributed database and provides transparency so users see it as a single logical database.

Q79. Sites of a geographically dispersed distributed database (for example, branches in different cities) are generally connected through a:

A. Personal Area Network (PAN) only

B. Single shared-memory bus

C. Standalone modem with no network

D. Wide Area Network (WAN)

Ans: D — Wide Area Network (WAN)

Explanation: Distributed databases rely on a communication network; widely separated sites are usually linked through a WAN.

--- End of Chapter 11.1 Question Bank ---